

APRD 6343

Applications of Advanced Statistical Techniques in Advertising

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (43.4 / 100)	36.8 / 100

37

CURRENT AVG SCORE
across 6 job roles

82

POTENTIAL SCORE
with recommended additions

+55

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Students develop advanced statistical models for digital advertising contexts, learning interpretation and client presentation skills.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	43.4	87.0	+43.6	HIGH
Machine Learning Engineer	40.1	78.5	+38.4	HIGH
Data Scientist	38.3	93.4	+55.1	HIGH
ML Engineer	37.0	79.3	+42.3	HIGH
Applied Scientist	35.9	85.2	+49.3	HIGH
MLOps Engineer	25.8	67.5	+41.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Deep Learning

2 of 6 roles

cnn

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2

MEDIUM

Consider Deep Learning content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.

GAP KEYWORDS BY ROLE

AI Engineer

43.4 → 87.0 (+43.6 pts)

data

robotics

software

technical

model

systems

detection

machine

Machine Learning Engineer

40.1 → 78.5 (+38.4 pts)

data

sets

tools

design

development

science

model

applications

Data Scientist

38.3 → 93.4 (+55.1 pts)

data

sets

different

social

approaches

ML Engineer

37.0 → 79.3 (+42.3 pts)

cnn

media

data

machine

systems

model

technical

building

Applied Scientist

35.9 → 85.2 (+49.3 pts)

research

reasoning

literature

technical

data

systems

industry

design

MLOps Engineer

25.8 → 67.5 (+41.7 pts)

detection

data

model

systems

machine

media

image

training

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.